

INDUSTRY WORK EXPERIENCE**NeuroErgonomics Labs, Madison, Wisconsin - Research Scientist | Technical Project Lead****Jan. 2025 – Current**

- Leading a multi-million dollar National Science Foundation-funded VR-based training initiative for emergency responders, integrating LLM-powered characters and usability testing to enhance training efficiency up to 25%.
 - Deployed large-scale data collection pipelines using Amazon Mechanical Turk (AWS) to gather behavioral and response data assessing user interactions with designed characteristics of LLM-powered characters.
 - Designed simulated law enforcement de-escalation training scenarios in VR using Unity, implemented through Meta Quest 3.
 - Identified gaps in existing emergency response protocols and formulated research hypotheses to guide the development of intelligent virtual training systems using AI-enabled feedback and real-time physiological data.
 - Designed evaluation of Human-AI interface based on context, literature and subject matter expert driven guidelines.
- Spearheading a multi-modal fatigue detection system to better understand and predict human states of fatigue and stress in individuals with high-stress jobs such as emergency medical service professionals.
 - Collaborated closely with stakeholders to integrate domain-specific insights into system design and usability evaluations.
 - Designed and conducted participant-facing studies to evaluate system usability.
- Developed end-to-end data processing pipelines for physiological, behavioral and contextual data.
 - Created statistical and inferential modeling workflows.
 - Built and validated machine learning models for prediction and forecasting of psychophysiological conditions.
- Translated data-driven insights into actionable system improvements.

Purdue University, Biomedical Engineering, West Lafayette, Indiana – Researcher (Doctoral Studies)**Jan. 2020 – Dec 2024**

- Created a non-invasive sensor-based diagnostic system for Parkinson's Disease tremor assessment, reducing inter-physician diagnosis variability and patient barriers through a bioinformatics and machine learning based telehealth solution.
- Developed a data-based method to analyze signals from commercially available wrist-worn inertial sensors to measure, distinguish and evaluate tremors that coexist in Parkinson's Disease patient populations.
- Engineered a data-driven tremor signal-based tremor evaluation metric which has shown greater resolution and better sensitivity than the current clinical gold standard in detecting the changes in tremor severity.
- Developed signal processing pipelines extract and denoise key tremor features from Inertial Measurement Unit (IMU) data.
- Designed statistical and machine learning workflows for detection, classification and prediction of severity of Parkinsonian symptoms.
- Collaborated with Parkinson's Foundation Center of Excellence and Indiana School of Medicine's Department of Neurology to align the diagnostic system with the current gold standard of clinical measures.
- Conducted structured interviews and iterative feedback sessions with neurologists and clinical researchers to refine diagnostic metrics for Parkinson's Disease severity assessment.
- Conducted extensive human subjects testing with Parkinson's patients and neurologists to assess the system's usability and diagnostic accuracy.
- Published and presented in several highly reputed journals and conferences such as Nature and Society for Neuroscience.

US Medical Innovations, Takoma Park, Maryland – Project Lead**Sept. 2016 – Dec. 2019**

- Led design and development of the Canady Robotics SR-70 Surgical System, an \$8M R&D program from concept to verification.
- Contributed to the launch the Canady Plasma XL-1000 Generator, the first to translate Cold Plasma technology from lab research to clinical use.
- Designed and prototyped mechanical components and PCBs, improving reliability, manufacturability, and signal integrity.
- Managed supplier communication to ensure compliance with ISO 13485 and medical-grade component standards.
- Defined user and safety requirements through use-scenario and task analyses, emphasizing intuitive and ergonomic design.
- Authored FDA design control documents including DHF, DMR, risk management (ISO 14971), and V&V reports.
- Supported regulatory testing under IEC 60601 (electrical safety), IEC 62366 (usability), and IEC 62304 (software lifecycle).
- Collaborated with clinicians, engineers, and regulatory teams to refine human-system interfaces and risk mitigations.
- Awarded 5 U.S. patents for innovations in electro-mechanical system design and medical device engineering.

Tupelo Brittany, Remote – Equity Consultant Intern**Jan. 2024 – June 2024**

- Delivered data-driven strategy engagements for biotech and pharmaceutical clients, synthesizing scientific literature, company filings (Form 10-K, investor reports), and market data into evidence-based recommendations for investment and portfolio decisions.
- Conducted sector and pipeline analyses connecting therapeutic innovation to competitive dynamics, valuation risk, and R&D investment trends; produced executive-ready summaries and visual presentations for senior stakeholders.

EDUCATION AND CERTIFICATES**Purdue University, West Lafayette, IN: Doctor of Philosophy in Biomedical Engineering****2020 – 2024**Research:

- Research focused on Parkinson's Disease to improve diagnostics by making it more objective and individual patient specific.
- Research showcased potential for up to 9% improvement in clinical diagnosis accuracy compared to the gold standard, utilizing an AI-based algorithm from wrist-worn sensor data.

Achievements:

- Awarded the title of Lamp Scholar in recognition of contributions to Weldon School of Biomedical Engineering, Purdue University.
- Awarded a \$100,000 PhD fellowship through the Computational Interdisciplinary Graduate Program for outstanding academic record.
- Awarded a \$75,000 PhD fellowship through the Leslie Bottorff Fellowship Program to adapt research into hospital and ambulatory care settings.

Purdue University, West Lafayette, IN: Master of Science in Mechanical Engineering**2016**Research:

- Developed an augmented reality based tele-mentoring system for thoracotomy surgery.
- Research was sponsored by the Office of the Assistant Secretary of Defense for Health Affairs.
- Research was reported in The Engineer, a London (UK) based magazine in 2015.

Achievements: Published in 5 papers in highly reputed journals.

CERTIFICATES

Purdue University, Krannert School of Management: Applied Management Principles

2020

Stanford University, Stanford Online: Machine Learning Certificate

2019

NCEES Certification: Fundamentals of Engineering

2014

TECHNICAL SKILLS

- **Programming:** MATLAB, Python, C++, CSharp, SQL.
- **ML/AI:** Scikit-learn, TensorFlow, PyTorch, signal processing, supervised/unsupervised learning, LLM, RAG.
- **Data:** Time-series analysis, Frequency-spectrum analysis, sensor fusion, feature engineering, statistical modeling.
- **Hardware/UX:** Wearables, haptics, VR/AR, human factors testing.
- **Tools:** SolidWorks, Autodesk Inventor, Altium Designer, Unity.

TEACHING WORK EXPERIENCE

Purdue University, Biomedical Engineering, West Lafayette, Indiana – Graduate Teaching Assistant

Aug. 2021 – Dec 2024

- Mentored, supported, and provided technical expertise to over 400 senior Biomedical Engineering undergraduates in the development of translatable medical devices and systems to address clinical needs.
- Empowered students with essential medical industry design and regulatory approval knowledge.
- Won the Estus H. and Vashti L. Magoon Award for outstanding instructor in 2023.

Purdue University, EPICS, West Lafayette, Indiana – Graduate Teaching Assistant

Aug. 2014 – May 2016

- Awarded Certificate in Practice of College Teaching from the Center for Instructional Excellence at Purdue University
- Supervised and supported 160+ undergraduates from various STEM departments in collaborative team projects, addressing engineering challenges to serve the local community and educational institutions.
- Instructed students from varying STEM disciplines in electromechanical systems design and fabrication, ethics, and user-centered design.

LEADERSHIP EXPERIENCE

- Co-founded the American India Foundation – Purdue Chapter: organized national level fundraising events to support social causes in India.
- Led a cross-functional team at US Medical Innovations, overseeing project goals and deadlines, technical R&D, and budget.
- Co-founded SPREE Technologies, a digital in-video advertisement startup.

RESEARCH PUBLICATIONS

- **Shanghavi, A. A.,** Jayasuriya, J., Cai, Y., & Mehta, R. K. (2025). Pulse Rate Variability From Wearable Wristwatches as a Surrogate for Heart Rate Variability? We Think Not Yet. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 0(0).
- Chen, W., Ruan, W., Pater, K., **Shanghavi, A.,** Mohanty, R., & Mehta, R. (2025, October 13–17). Human factors evaluation of Gen-AI based NPC interactions in public safety training. ASPIRE 2025: 69th International Annual Meeting of the Human Factors and Ergonomics Society, Chicago, IL.
- **A Shanghavi,** R Patil, B S. Duerstock, S Ambike and A B. Sereno. A data-driven method for detecting, differentiating and scoring Parkinsonian rest and postural tremors. npj Parkinsons Disease. Submitted
- **A Shanghavi,** R Patil, B S. Duerstock, S Ambike and A B. Sereno. A machine-learning method for detecting simulated Parkinsonian tremors in healthy individuals. Advanced Intelligent Systems. Submitted.
- **A Shanghavi,** D Larranaga, R Patil, E M. Frazier, S Ambike, B S. Duerstock and A B. Sereno. A machine-learning method isolating changes in wrist kinematics that identify age-related changes in arm movement. Scientific Reports 14, 9765 (2024). <https://doi.org/10.1038/s41598-024-60286-1>
- **A. Shanghavi,** R. Patil, J. Stad, G. Maffei, B. S. Duerstock, A. B. Sereno. Power Spectral Density Analysis of Kinematic Variables Reveals Age-related Differences in Arm Motion. Program No. PST151.13/BB14. 2023 Neuroscience Meeting Planner. Washington, D.C.: Society for Neuroscience, 2023. Online
- Sakaleros, J., Shamloo, F., **Shanghavi, A.,** & Sereno, A. (2021). Detection of Parkinsonian Motor Symptoms Using Non-Invasive Motion Sensing Technology. Proceedings of IMPRS, 4(1).
- Andersen, D.; Popescu, V.; Cabrera, M.; **Shanghavi, A.;** Gomez, G.; Marley, S.; Mullis, B. & Wachs, J. (2016). Medical Telementoring Using an Augmented Reality Transparent Display. Surgery, 159(6), 1646 – 1653.
- Andersen, D.; Popescu, V.; Cabrera, M.; **Shanghavi, A.;** Gomez, G.; Marley, S.; Mullis, B. & Wachs, J. Avoiding Focus Shifts in Surgical Telementoring Using an Augmented Reality Transparent Display. NextMed/MMVR22 Proceedings, 2016.
- Andersen, D.; Popescu, V.; Cabrera, M.; **Shanghavi, A.;** Gomez, G.; Marley, S.; Mullis, B. & Wachs, J. Virtual annotations of the surgical field through an augmented reality transparent display. The Visual Computer, Springer Berlin Heidelberg, 2015, 1-18.
- Dan Andersen, Voicu Popescu, Maria Eugenia Cabrera, **Aditya Shanghavi,** Brian Mullis, Sherri Marley, Gerardo Gomez, Juan P. Wachs; An Augmented Reality-Based Approach for Surgical Telementoring in Austere Environments, Military Medicine, Volume 182, Issue suppl_1, 1 March 2017, Pages 310–315
- Andersen, D., Popescu, V., Lin, C., Cabrera, M. E., Shanghavi, A., & Wachs, J. (2016). A Hand-Held, Self-Contained Simulated Transparent Display. 2016 IEEE International Symposium on Mixed and Augmented Reality (ISMAR-Adjunct).

PATENTS

- Canady, J., O'connell, M., Yan, F., Liu, Z., & **Shanghavi, A.** (2020). U.S. Patent Application No. 29/644,182.
- Canady, J., Canady, C., Zhuang, T., Yan, F., **Shanghavi, A.,** Liu, Z., ... & Pierce, T. (2022). U.S. Patent No. 11,224,487. Washington, DC: U.S. Patent and Trademark Office.
- Canady, J., O'connell, M., Yan, F., Liu, Z., & **Shanghavi, A.** (2020). U.S. Patent Application No. 29/644,181.
- Canady, J., Yan, F., Zhuang, T., **Shanghavi, A.,** Ray, L., Sumanasena, B., & Sokolovski, E. (2023). U.S. Patent Application No. 18/126,799.
- Canady, J., Yan, F., Zhuang, T., **Shanghavi, A.,** Ray, L., Sumanasena, B., & Sokolovski, E. (2023). U.S. Patent No. 11,612,425. Washington, DC: U.S. Patent and Trademark Office.